

Signals and Systems Study Notes (EC301)

Academic Insight - Public Sample Revision Summary

Module 1: LTI Systems & Convolution

- Continuous convolution: $y(t) = \int x(\tau) h(t - \tau) d\tau$
- Discrete convolution: $y[n] = \sum x[k] h[n - k]$, length = $N1 + N2 - 1$
- Causal LTI: $h(t) = 0$ for $t < 0$; BIBO stable: $\int |h(t)| dt < \infty$

Module 2: Fourier, Laplace & Nyquist Sampling

- Convolution in time $\leftrightarrow$ Multiplication in frequency: $Y(j\omega) = X(j\omega) H(j\omega)$
- Nyquist Sampling Criterion: $f_s \geq 2 * f_{\max}$ to prevent aliasing
- Note: Sample educational revision sheet provided by Academic Insight.